

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)
(Estd. under the Panjab University Act VII of 1947 –enacted by the Govt. ofIndia)

SYLLABI

Course Outline for
B.Voc (SOFTWARE AND DEVELOPMENT)
as per NEP 2020

w.e.f session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
SD-101	Core Discipline	Basic Computer Skills	4	--	4	80	20	--	--	100
SD-102	Core Discipline	Logic Development Technique	2	--	4	40	10	--	--	100
SD-102P		Practical Based on SD- 102	--	2		--	--	40	10	
SD-103	Core Discipline	Fundamental of Information & Technology II	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
SD-201	Core Discipline	Relational Database Management System	4	--	4	80	20	--	--	100
SD-202	Core Discipline	Programming in C Language	2	--	4	40	10	--	--	100
SD-202P		Practical Based on SD-202	--	2		--	--	40	10	
SD-203	Core Discipline	RDBMS using MySQL	--	4	4	--	--	80	0	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[2]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[1]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	--	4	4	--	--	80	20	100
Total Credits					**24**					**600**

Job Role - Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

*Summer Industrial Training of 2-4weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be heldwithin 3-weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in (Software and Development) = 48 + 4 #.
#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.
***One credit of theory lecture is of 1 hour duration and one credit of practical is of 2 hours duration.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF
B.Voc. (SOFTWARE AND DEVELOPMENT)

PROGRAMME OUTCOMES

PO1. Acquire various skills in the field of Information Technology. This includes skills related to various office automation tools like MS Office, PowerPoint, MS Excel, etc., and their applications.

PO2. Gain knowledge about the fundamentals of computer systems and their working. Further students will be able to develop programming skills with programming languages C, JAVA, Python, etc. by actually designing programs in concerned laboratories.

PO3. Learn about computer networking, operating systems, and related areas both theoretically and with practice by performing experiments in the laboratories.

PO4. Design, develop, and implement data structures and algorithms for problem-solving in various domains by virtue of training in the laboratories.

PO5. Learn scripting and web development tools like PHP, JavaScript, etc., and apply these to design and develop real-world tools.

PO6. Equip themselves with the knowledge of financial accountancy and develop expertise in software like Tally.

PO7. Get exposure to the modern concepts of computing like Artificial Intelligence, Data Science, E-Commerce, and various Internet Technologies as part of this Programme.

PO8. Get industrial exposure in the field of information technology through an internship with industry.

PO9. Seek employment in various jobs in the Government sector as well as IT and related industries and perform various roles related to software development, testing, and maintenance.

SEMESTER I

	COURSE TITLE	BASIC COMPUTER SKILLS
1.	**Course Code**	SD-101
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	60 hours of lectures for Theory credits =4 (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100 Marks (80TH+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Field Technician- Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician
7.	**Course Objectives**	The course aims to equip students with a profound understanding of computer fundamentals, hardware, software, and networking. To empowers the ability to analyze advantages and disadvantages, design computer system architectures, apply input and output devices, and comprehend emerging technologies and well-roundedexpertise in basics concepts.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1Demonstrate a comprehensive understanding of computer fundamentals, including the historical development and characteristics of computers. CO2Apply knowledge of input and output devices, computer memory, and different types of software in practical scenarios. CO3Analyze and evaluate the functions and types of operating systems, data processing systems, and translators. CO4Demonstrate proficiency in understanding computer networks, the Internet, and emerging technologies, with the ability to classify and describe differenttypes of networks.
9.	**Pedagogy of the Course Work**	Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Computers: Characteristics, Historical development of computers, Evolution of computing devices, Generations of Computers: First Generation (Vacuum Tubes, Batch Processing), Second Generation (Transistors), Third Generation (Integrated Circuits, Time-Sharing Systems), Fourth Generation (Microprocessors, Personal computers), Fifth Generation (Artificial intelligence). Hardware, Software and Firmware, Data and Information. Computer Codes: Decimal System, Binary System, Hexadecimal System, Octal System and ASCII Code. Decimal to Binary conversion, Binary to Decimal Conversion.

UNIT– II

Introduction to Computer Networks: Definition and Importance of Computer Networks, Basic Components of a Computer Network, Classification of Networks, Network Topologies. Internet: Historical Overview of The Internet, Evolution of The World Wide Web (WWW), WWW Components, Internet Governance and Standards, Internet Services and Applications (Email Protocols, Telnet, FTP, IRC, and Usenet News). Internet Access Methods (Dial-Up, DSL, Cable, Fiber-Optic), Browser Technologies and Features.

UNIT– III

Computer Virus: Types of Viruses, Impact of Virus, Virus Detection and Prevention techniques, Malware. Antivirus and Its Types. Multimedia: Hardware and Software for Multimedia, Multimedia: Component of Multimedia, Basic File Formats, Audio File Formats, Image File Formats, Video File Formats. Multimedia Storage Devices.

UNIT-IV

Emerging Technologies: Basics of Internet of Things (Iota), Components of Iota, Applications of Iota, Basics of Parallel Computing and Distributed Computing, Difference between Parallel computing and Distributed Computing, Advantages and Limitations of Parallel computing and Distributed Computing, Cloud Computing: Component of Cloud Computing, Types of Cloud Computing, and their applications.

13. Suggested Resources for Reading

1. Fundamentals of Computers, V. Rajaraman, PHI Publication.
2. Computer Fundamentals, P. K. Sinha, BPB Publication.
3. Computers Today, Basandra, S. K., Galgotia Publications.
4. Fundamentals of Information Technology, Anshuman Sharma, Lakhanpal Publications.
5. PC Hardware: The Complete Reference, Craig Zacker., TMH Publication.

Note: Latest edition of text books may be used.

	COURSE TITLE	LOGIC DEVELOPMENT TECHNIQUE
1.	**Course Code**	SD-102
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	30 hours of lectures for Theory + 60 hours of practical'sTheory credits =2 and Practical credits=2 (1 credit = 15 hours of theory learning based on 2 hrs./week) +(1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks ((40TH+10IA) +(40PR+10IA)) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer
7.	**Course Objectives**	This course aims to provide comprehensive knowledge of computer peripherals, hardware components, and system management. Hardware Fundamentals enable to learn about hardware technologies, their implementation/installation, uses, troubleshooting and maintenance.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. To develop the logic of the different problems. CO2. To equip students with the basic knowledge of logic development techniques. CO3. To enhance the skills of students for software development.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and GroupDiscussions etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT-I

Logic Development Tools**:** Algorithm Development: Types of Algorithms, Advantage and Disadvantage of Algorithm, Analysis of Algorithms, Complexity of Algorithm, Big-O Notation. Flowchart**:** Types of Flowcharts, Advantage and Disadvantage of Flowchart, Designing Pseudo code, Decision table, Steps for problem solving, Stepwise refinement, Modular Programming: top down & bottom-up approach.

UNIT-II

System Analysis and Design: Definition of problem, Feasibility study, Requirement analysis, Designing, Coding, Testing, and Implementation of software project.

UNIT-III

Basics of 'C' Programming Language Usage of Character Set, Meaning of Keywords and Identifiers, Role of Data Types, Constants and Variables. Importance of type Casting. Different Types of Operators and their Precedence, Expressions, Conditional Statements

UNIT-IV

Control Statements: Decision Making statements (if, if-else, else if, nested if), Looping Statements (For, While, do-while), Usage of Exit, Continue, Break and Goto Statement. Introduction of array and functions

12. Suggested Resources for Reading

1. Kanetkar, Y.P., Let us C, BPB Publications
2. Balagurusami, E., Programming in ANSI C, Fourth Edition, Tata McGraw Hill
3. Sharma, A., Learn Programming in C, Lakhanpal Publisher
4. Kernighan, B.W., & Richie, D.M., The C Programming Language, PHI Publication
5. R. S. Salaria, Problem Solving and Programming in C, Khanna Publishing Pvt. Ltd.
6. Awad, E. M., System Analysis and Design, Galgotia Publication Pvt. Ltd.

Note: Latest edition of text books may be used.

SD-102P (Practical Based on SD-102)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

COURSE TITLE	FUNDAMENTALS OF INFORMATION TECHNOLOGY II
1. Course Code	SD-103
2. Typology of Course	Core Discipline
3. Course Duration	120 hours of practical's credits=4 (1 credit = 30 hours of practical based on 8 hrs./week)(1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit
4. Credits	4 Credits
5. Max. Marks and Time	100Marks (80PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6. Job Role	Field Technician- Computing & Peripherals
7. Course Objectives	The course aims to provide a comprehensive understanding of computer systems, covering fundamental hardware components. It develops practical skills in connecting peripherals, analyze and solve hardware issues logically, and assemble, configure, troubleshoot, and maintain functional computer systems. Additionally, the course focuses on installing Windows, managing drivers, and implementing antivirus software for system security.
8. Course Outcomes	On successful completion of the course, the students will be able to: CO1. Define and explain the fundamental components of a computer system, demonstrating a holistic understanding of hardware architecture. CO2. Demonstrate practical skills in connecting and configuring peripherals, including mouse, keyboard, display devices, scanning devices, and hard disk. CO3. Develop critical thinking skills to analyze and solve hardware-related problems, applying a logical and systematic approach to diagnose and address issues. CO4. Assemble, configure, troubleshoot, and maintain a functional computer system, integrating knowledge from all units for practical application. CO5. Successfully install the Windows operating system, manage the installation and updating of drivers, install and update anti-virus software.
9. Pedagogy of the Course Work	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
10. Instructions for paper setters and candidates	i) Examiner will set total of four questions covering the whole syllabus. ii) Student will attempt any three questions. iii) Each question carries 10 marks. iv) Practical file carries 10 marks and 40 marks for practical work and viva-voce.

11. Course Content

UNIT-I

Operating Systems- Windows and Unix. Fundamentals of windows, anatomy of windows, Operations on window: Opening, Minimizing, Maximizing, Moving, Resizing, closing; Windows Explorer, Folders: Creating, deleting, copying, renaming folders, folder properties; Icons, Menu, Taskbar, Control panel, Recycle bin. Overview of UNIX structure, general purpose UNIX commands such as date, echo, call, be, pawed, passed; file and directory commands such as less, midair, cap, mv, rm. process management commands such as PS, kill, no up, communication commands such as news, mess, wall; working with editor introduction to shell programming

UNIT-II

Word Processing Package: Basics of Word Processing; Word Processing Basics; Text creation, Manipulation, Finding and replacing, Formatting of text; Printing of word document, Page Layout: Margin setting, Alignments, Adding Borders and shading, Adding Headers and Footers, setting up Multiple columns, working with tables, Spell check, Auto Correct, Grammar facility, Retrieving often used text; Auto text character formatting, language setting and thesaurus; Macros; Mail merge.

UNIT-III

Spreadsheet Package: Worksheet Basics, Data Entry in Cells: Entry of numbers, text and formulae, moving data in a worksheet, moving around in a worksheet, Selecting Data Range. Using the interface (Toolbars, Menus), Editing Basics, working with workbooks, Cell referencing: Absolute, Relative and Mixed; Formatting and Calculations: using Auto fill, Working with Formulae, Efficient Data Display with Data formatting (number formatting, date formatting etc.), Function: different types of functions and their usage Worksheet Printing. Working with Graphs and Charts: Creating Chart and graphs, using chart wizard, sizing and moving parts, updating charts, Changing chart types, Formatting Charts. Database Management; Finding records with Data form, Adding/Deleting Records, Filtering Records in a worksheet, Sorting; Macros: Creating Macros, Record Macros, Running Macros, Assigning Macros to Buttons.

UNIT-IV

Presentation Packages: Basics, General Features, Creating a presentation, Different types of slide views, Master Slides and its use, Formatting Slides: slide design, Layout and background; Animation effect, Transition effect, timing effects, Macros.

Database Package: Introduction to Database, Tables, Data Types, Attributes, Records; Overview of MS-ACCESS, Creating Database, Creating Tables, Data types, Importing and Exporting data, using Wizards, Creating forms and queries.

12. Suggested Resources for Reading

1. Sinha, P.K., Computer Fundamentals, BPB Publications
2. Rajaraman, V., Fundamentals of Computers, PHI
3. Mathur, R., DOS 6.2 Quick Reference, Galgotia Publication's Pvt. Ltd.
4. Sharma, A., Fundamentals of Information Technology, Lakhanpal Publisher
5. Texel, R.K.: PC Software Made Simple, Tata McGraw Hill.

Note: Latest edition of text books may be used.

SEMESTER II

	COURSE TITLE	RELATIONAL DATABASE MANAGEMENT SYSTEM
1.	**Course Code**	SD-201
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	60 hours of lectures for Theory Theory credits =4 (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks (80TH+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the ExaminationHall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer
7.	**Course Objectives**	i) To provide good understanding of database design, modelling and implementation of database management systems. ii) To educate students with the fundamental concepts of Data Base Design, Data Models and different Data Base Languages. iii) To make access to the data easy for the user.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. Understand the basic concepts of database management systems. CO2. Apply SQL to find solutions to a broad range of queries. CO3. Apply normalization techniques to improve database design. CO4. Analyze a given database application scenario to use ER model for conceptual design of the database.
9.	**Pedagogy of the Course Work**	Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT-I

Introduction: Database vs. File Oriented Approach, Advantages and Disadvantages of DBMS, Data models: Hierarchical, Network and Relational models.

UNIT - II

Entity Relationship model, Client-Server Architecture, Data Independence. Concept of Relational Algebra and Calculus, Storage Organization for Relations

UNIT-III

Relational Algebra: Operations - union, intersection, difference, Cartesian product, projection, selection, division and relational algebra queries; Relational Calculus: Tuple oriented and domain oriented relational calculus and its operations, Normalization and its forms. DBMS vs RDBMS

UNIT-IV

Database Integrity, Security, Concurrency, Backup and Recovery of the database. Design and structure of Distributed databases

12 Suggested Resources for Reading

1. Desia, B.C., An Introduction to Database System, Galgotia Publications,
2. Date, C.J. & Sharma, A., Database System Vol I & II, Narosa Publications
3. Sharma, A., Fundamentals of DBMS, Lakhanpal Publisher
4. Korth, H.F., Database System Concepts, McGraw Hill

Note: Latest edition of text books may be used.

	COURSE TITLE	PROGRAMMING IN C LANGUAGE
1.	**Course Code**	SD-202
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	30 hours of lectures for Theory + 60 hours of practicals Theory credits =2 and Practical credits=2 (1 credit = 15 hours of theory learning based on 2 hrs./week) + (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks ((40TH+10IA) +(40PR+10IA)) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the ExaminationHall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer
7.	**Course Objectives**	i) To help the students in finding solutions to the various real-life problems and converting the solutions into computer program using C. ii) To acquire knowledge about the basic concept of writing a program. iii) To learn and acquire the art of computer programming.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. Develop a C program CO2. Control the sequence of the program and give logical outputs CO3. Implement strings in your C program CO4. Store different data types in the same memory CO5. Manage I/O operations in your C program CO6. Repeat the sequence of instructions and points for a memory location CO7. Apply code reusability with functions and pointers CO8. Understand the basics of file handling mechanisms CO9. Explain the uses of pre-processors and various memory models
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and GroupDiscussions etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT-I

Fundamental of 'C': Problem Solving steps, Structure of C Program, I/O statements, assignment statements, C character Set, Constants, Variables, Operators and Expressions, Standards and Formatted statements, Keywords, Data Types and Identifiers.

Control Structures: Introduction, Decision making with if – statement, if-else and nested if, While and do-while, for loop. Jump statements: break, continue, go to, switch Statement.

UNIT-II

Functions: Function Declaration and Definition, types of functions, Call by value and Call by reference, Recursion, Pre-processor Directives.

UNIT-III

Preprocessor Directives: Introduction and Use, Macros, Conditional Preprocessors and Header Files.

Arrays: Introduction, Types, Operations on Arrays.

UNIT-IV

Structure and Union: Declaring structure, accessing structure members, Array of structure, nested structure, Unions.

Files**:** introduction, creating, opening of data files, closing a data file and file handling functions

12. Suggested Resources for Reading

1. Kanetkar, Y.P., Let us C, BPB Publications
2. Balagurusami, E., Programming in ANSI C, Fourth Edition, Tata McGraw Hill
3. Gottfried, B.S., Programming in C, McGraw Hills.
4. Sharma, A., Learn Programming in C, Lakhanpal Publisher
5. R. S. Salaria, Problem Solving and Programming in C, Second Edition

Note: Latest edition of text books may be used.

SD-202P (Practical Based on SD-202)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

	COURSE TITLE	RDBMS USING MYSQL
1.	**Course Code**	SD-203
2.	**Typology of Course**	Core Discipline
3.	**Course Duration**	120 hours of practicals Practical credits=4 (1 credit = 30 hours of practical based on 8hrs./week) (1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks (80PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer
7	**Course Objectives**	i) To introduce principles and foundations of databases, including architecture, design issues, integrity control, query processing and optimization, transactions, and concurrency control. ii) To design and implement database for enterprise application.
8	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. Understanding Database Concepts CO2. Understanding Database Storage & Database Security CO3. Understanding Entities and Relationships & Normalization CO4. Understanding the Relational Data Model CO5. Understanding Database Design and Performance Tuning CO6. Understanding Database Backup and Restore & MySQL
9	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
10	**Instructions for paper setters and candidates**	i) Examiner will set total of four questions covering the whole syllabus. ii) Student will attempt any three questions. iii) Each question carries 10 marks. iv) Practical file carries 10 marks and 40 marks for practical work and viva-voce.

11. Course Content

UNIT-I

Database Fundamentals: What is a database? What is MySQL? Introduction to MySQL & MySQL workbench.

Database Design: Introduction To Database Design, Learn Database Normalization with the help of a case study, Learn ER Modelling with a Case Study

UNIT-II

MySQL Basics: Creating a Database & MySQL Data Types, SELECT in MySQL, how to query data using Where clause in MySQL, Insert Into in MySQL, Delete & Update data in MySQL

Data Sorting: How sorting is done in MySQL, using ORDER BY, DESC and ASC, Group By in MySQL, Wildcards in MySQL, Using Regular Expressions & Wild Cards in MySQL

UNIT-III

Functions: Introduction to functions, Aggregate Function in MySQL, Null value & Keyword in MySQL, Auto Increment in MySQL, Alter, Drop & Rename function in MySQL, Limit keyword in MySQL

UNIT-IV

Queries: Using Sub-Queries in MySQL, joins in MySQL, Unions in MySQL, Views in MySQL, Index in MySQL Minor Application Design: Creating an application using MySQL

12 Suggested Resources for Reading;

1. Desai, B.C., An Introduction to Database System, Galgotia Publications.
2. Korth, H.F., Database System Concepts, McGraw Hill
3. Date, C.J., Kannan, A. & Swamynathan, S., An Introduction to Database system, Pearson Publications
4. Jain, M., Pillai, V., Singh, S. & Jain,S., Introduction to Database Management, BPB.
5. Schwartz, B., & Zaitsev, P., High Performance MySQL: Optimization, Backups, Replication, and More.
6. Dyer, R.J.T., MySQL in a Nutshell, O'Reilly.

Note: Latest edition of text books may be used.